AWS NAT Gateway: Enabling Internet Connectivity for a Private Subnet

A Step-by-Step Practical Guide to VPC Networking, Route Tables, EC2 Connectivity, ICMP Testing, and NAT Gateway Configuration

PRACTICAL LAB REPORT

AWS networking · IPv4 · Mumbai Region (`ap-south-1`)

IPv4 routing, EC2, and outbound connectivity

Reproducible configuration guide with evidence limitations documented

1 Introduction

Cloud workloads often have different connectivity needs. An instance in a public subnet can communicate directly with the internet when its subnet has an Internet Gateway (IGW) route, the instance has public IPv4 connectivity, and traffic is allowed. A private-subnet instance is usually kept without a public IPv4 address, but may still need outbound access for updates, package downloads, or external services.

A public NAT Gateway provides a managed path for eligible private IPv4 resources to initiate outbound connections without assigning those instances public IPv4 addresses. This lab asks: *Why could the public EC2 instance ping Google's public DNS address before NAT was configured while the private instance could not, and what changed when NAT Gateway was introduced?*

This report combines the learner's supplied configuration and reported outcomes with the console screenshots named in the request. Those screenshots are not present in the current workspace. Each intended evidence location is explicitly marked by its expected filename; no screenshot or terminal output has been synthesized. Add the original PNGs beside this source file and recompile to insert them automatically.

2 Objectives

- Create a custom VPC and plan public and private IPv4 subnets.
- Understand CIDR addressing, subnet boundaries, and address capacity.
- Create and attach an Internet Gateway; configure public and private route tables.
- Configure security groups and launch public and private EC2 instances.
- Test ICMP connectivity to `8.8.8.8` before and after NAT.
- Create a public NAT Gateway and route private-subnet internet-bound traffic through it.
- Compare observed connectivity, interpret the routing change, and review cleanup and cost considerations.

3 Prerequisites

You need an AWS account and an appropriately permissioned IAM user. These foundations are sufficient:

- A **VPC** is an isolated virtual network in an AWS Region. A **subnet** is a portion of its IP space in one Availability Zone (AZ).
- An IPv4 **CIDR** block expresses an address range; `10.0.1.0/24` contains 256 addresses before AWS subnet reservations.
- A **route table** selects a next hop for traffic by destination. `0.0.0.0/0` is an IPv4 default route.
- An **Internet Gateway** connects a VPC to the internet when routing, addressing, and traffic controls permit it. A **NAT Gateway** allows private IPv4 resources to initiate outbound internet connections.
- A **security group** is a stateful virtual firewall attached to network interfaces. EC2 instances are virtual servers.
- **ICMP Echo Request/Echo Reply** messages are the request and response used by IPv4 `ping`; ICMP does not use TCP or UDP ports.

4 Fundamental concepts

The terms *public* and *private* describe network reachability, not a subnet checkbox. A subnet is effectively public when its associated route table sends internet-bound traffic to an IGW. A public IPv4 address is also needed for direct IPv4 communication with the internet. A private subnet has no direct IGW route; its instances can remain without public IPv4 addresses.

Route tables and security groups have different jobs: a route table chooses the next hop, while a security group allows or denies traffic to and from an instance. NAT Gateway is a managed translation and forwarding service, not a general-purpose inbound firewall or a way to initiate arbitrary unsolicited internet connections to private instances.

5 IAM preparation

The learner created a custom `VPC-IAM` policy and attached it alongside the visible `EC2-IAM`, `S3-IAM`, and `AmazonEC2ReadOnlyAccess` permissions. Routine administrative work should use a permissioned IAM identity rather than the root user, reserving root for tasks that require it. IAM controls which AWS API actions an identity may perform; it does not grant an EC2 instance network connectivity.

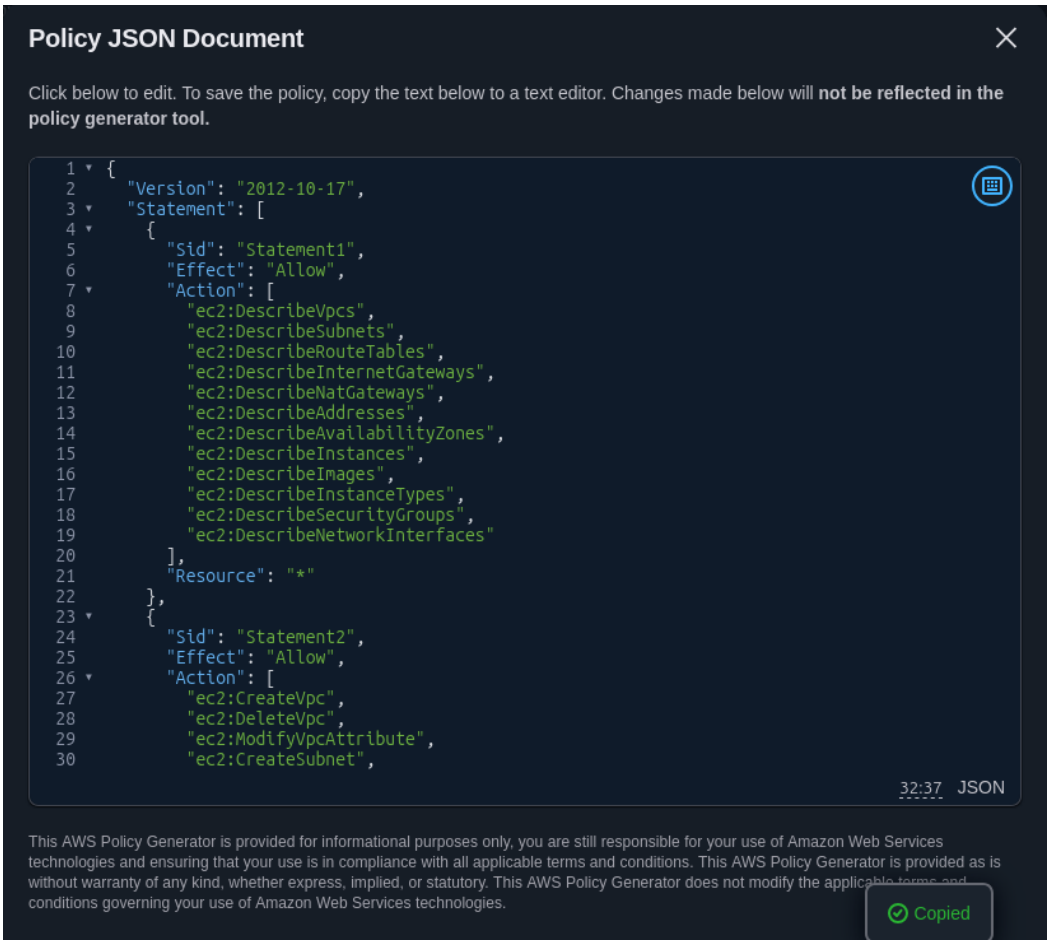


Figure 1: Custom VPC IAM policy creation.
The supplied filename is intended to show creation of the custom `VPC-IAM` policy. It does not establish every policy permission or prove production-ready least privilege.

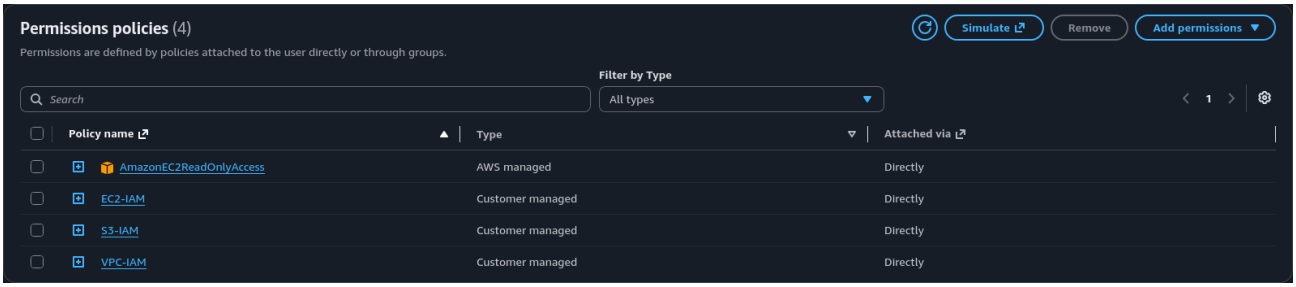


Figure 2: IAM policy attachments visible for the lab user.
The screenshot is described as showing `EC2-IAM`, `VPC-IAM`, `S3-IAM`, and `AmazonEC2ReadOnlyAccess`. Account identifiers are intentionally not reproduced.

6 Network architecture and addressing plan

The lab uses a custom VPC in Asia Pacific (Mumbai), Region `ap-south-1`. Both subnets were created in the same AZ according to the supplied subnet screenshots. The VPC uses a larger address range so it can contain smaller, non-overlapping subnets.

Resource	Name	IPv4 CIDR or route
VPC	`NAT-Lab-VPC`	`10.0.0.0/16`
Public subnet	`NAT-Public-Subnet`	`10.0.1.0/24`
Private subnet	`NAT-Private-Subnet`	`10.0.2.0/24`
Internet Gateway	`NAT-Lab-IGW`	Attached to VPC
Public route table	`NAT-Public-RT`	Default route to IGW
Private route table	`NAT-Private-RT`	Initially local only; later default route to NAT
Public security group	`NAT-Public-SG`	SSH from learner IP; outbound allow-all
Private security group	`NAT-Private-SG`	SSH from public SG; outbound allow-all
NAT Gateway	`NAT-Lab-Gateway`	Public NAT Gateway
Public EC2	`NAT-Public-EC2`	In public subnet
Private EC2	`NAT-Private-EC2`	In private subnet; no public IPv4

The VPC range `10.0.0.0/16` spans `10.0.0.0` through `10.0.255.255`. Public subnet `10.0.1.0/24` spans `10.0.1.0` through `10.0.1.255`; private subnet `10.0.2.0/24` spans `10.0.2.0` through `10.0.2.255`. Both /24 blocks fit in the /16 and do not overlap. AWS reserves five IPv4 addresses in each ordinary subnet, leaving 251 assignable addresses per /24. This is subnet capacity, not a required EC2 instance count.

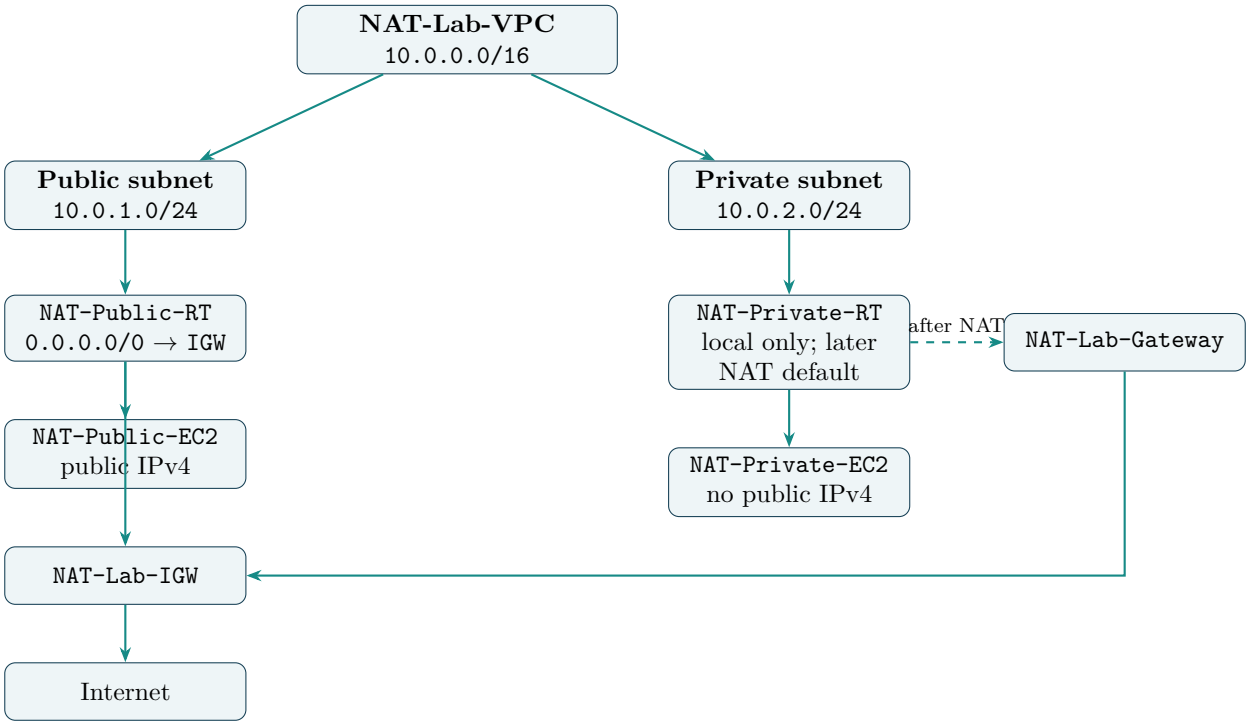


Figure 3: Logical lab topology. Arrows indicate routing relationships rather than physical devices. The private default route is added only after NAT is available.

7 VPC creation

The learner selected VPC-only creation, named the VPC `NAT-Lab-VPC`, and set its IPv4 CIDR to `10.0.0.0/16`. A custom VPC keeps the lab's address space and routes explicit instead of relying on default-VPC settings or unrelated resources. VPC creation alone does not create internet access; subnets, gateways, routes, addresses, and traffic controls are configured separately.

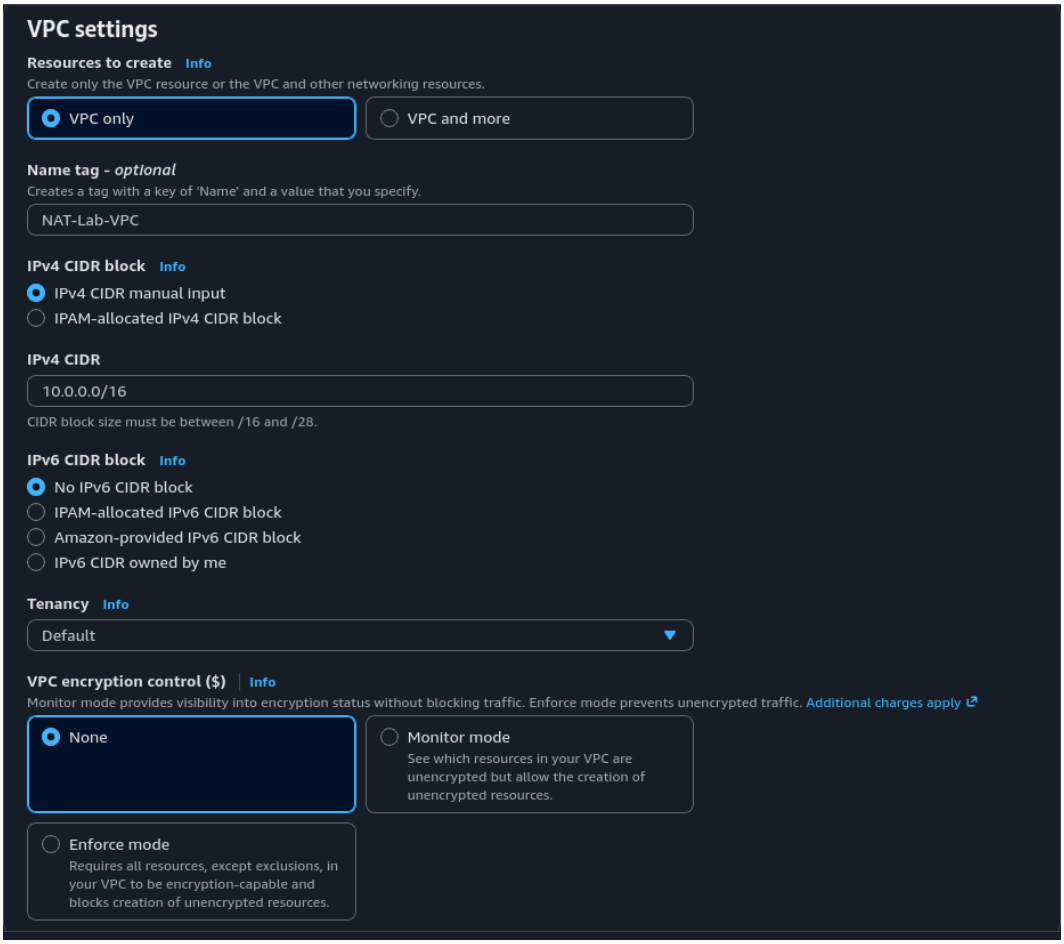


Figure 4: Custom VPC configuration: `NAT-Lab-VPC`, `10.0.0.0/16`.
Notice the VPC-only workflow and planned CIDR. A creation form documents configuration, not by itself successful creation.

8 Public and private subnet creation

Create `NAT-Public-Subnet` with `10.0.1.0/24` and `NAT-Private-Subnet` with `10.0.2.0/24`. Select the intended VPC and AZ. Using the same AZ simplifies this single-zone experiment and avoids making cross-AZ availability or charges part of the comparison; a production design may span AZs for resilience.

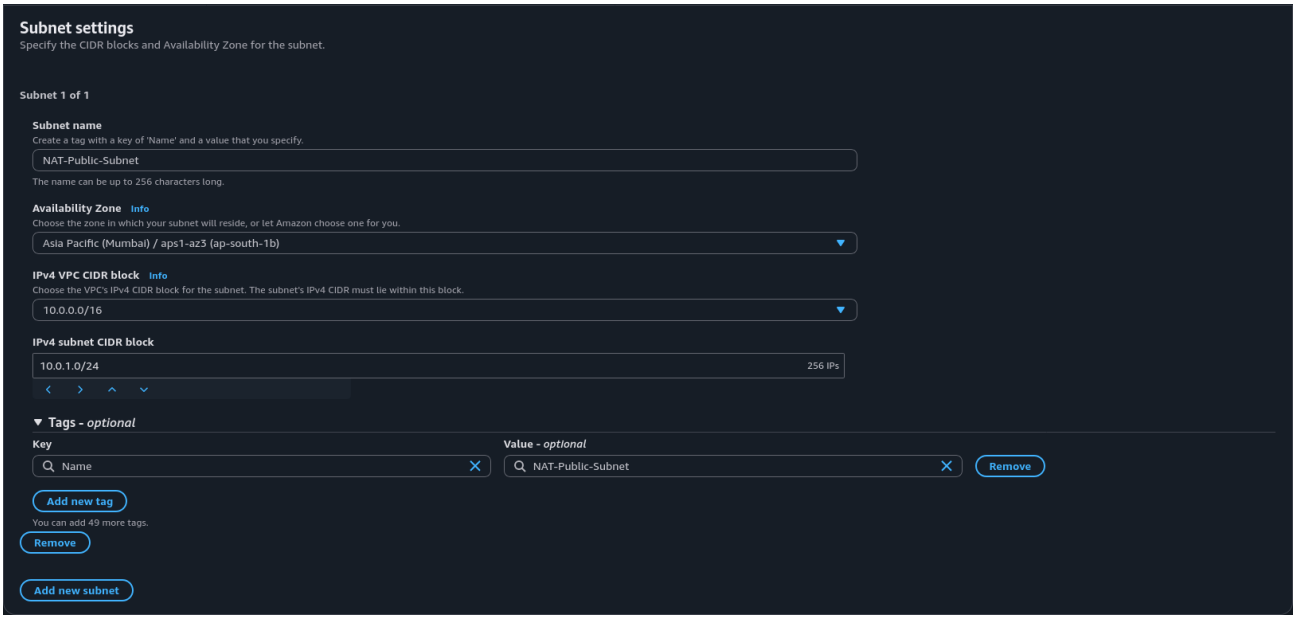


Figure 5: Public subnet configuration using `10.0.1.0/24`.
Check subnet name, VPC, CIDR, and intended AZ.

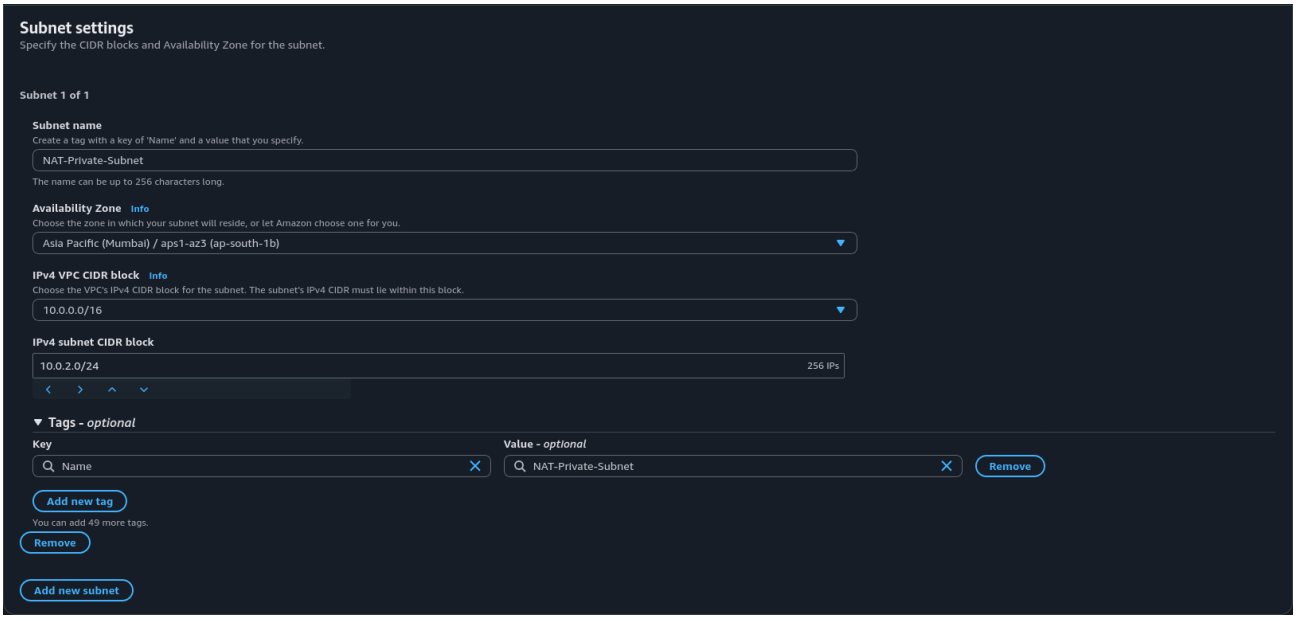


Figure 6: Private subnet configuration using `10.0.2.0/24`.
Check that the CIDR is within the VPC, non-overlapping, and in the selected AZ for this lab.

Automatic public IPv4 assignment was enabled for the public subnet. This setting can automatically associate public IPv4 addresses with launched instances, but does not by itself make a subnet public. Its route table also needs a route to an IGW, with suitable addressing and traffic controls.

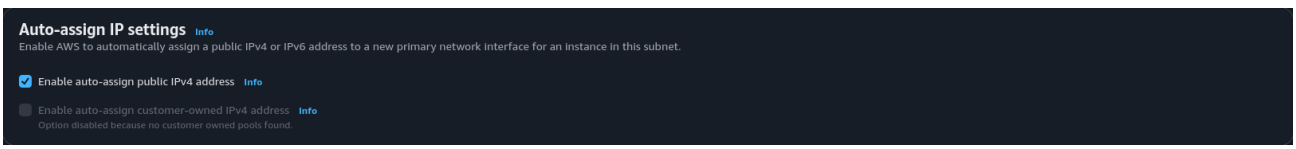


Figure 7: Automatic public IPv4 assignment for the public subnet.
This supports public addressing at launch; routing through an Internet Gateway is also necessary for a public subnet.

9 Internet Gateway creation and attachment

Create `NAT-Lab-IGW` and attach it to `NAT-Lab-VPC`. An IGW attaches to a VPC, not directly to a subnet or instance. It provides a target for internet-bound routes but does not automatically make every resource internet-accessible. Routes, public addressing, security groups, network ACLs, and instance configuration also matter.

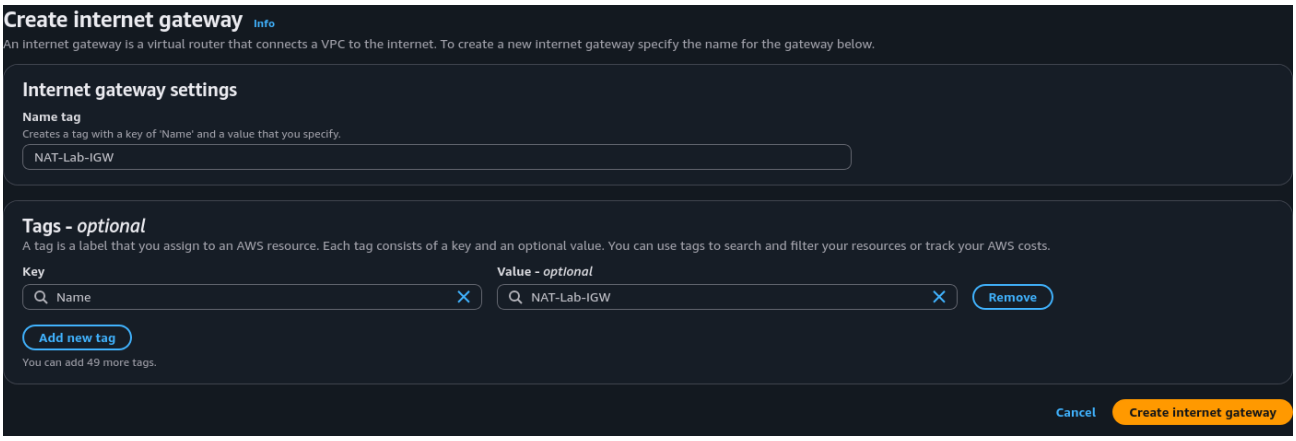


Figure 8: Internet Gateway creation for `NAT-Lab-IGW`.
The IGW is a VPC routing target; creation is distinct from attaching it.

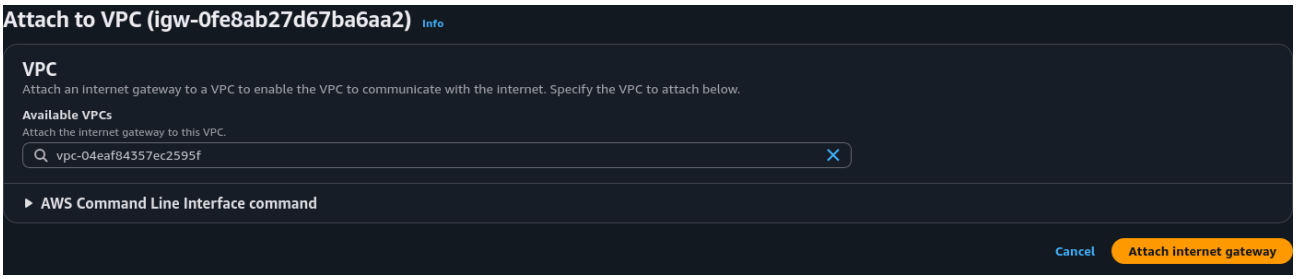


Figure 9: Attaching `NAT-Lab-IGW` to `NAT-Lab-VPC`.
Subnets use the IGW indirectly through their associated route tables.

10 Route table configuration

10.1 Public route table

Create `NAT-Public-RT`. Its VPC-local route is separate from its internet default route:

Destination	Target	Purpose
`10.0.0.0/16`	`local`	Routes within the VPC
`0.0.0.0/0`	`NAT-Lab-IGW`	Other IPv4 destinations toward the IGW

Associate `NAT-Public-Subnet` with this table. This explicit association makes the public subnet use the IGW default route. The supplied association screenshot documents the public subnet only, not the private-subnet association.

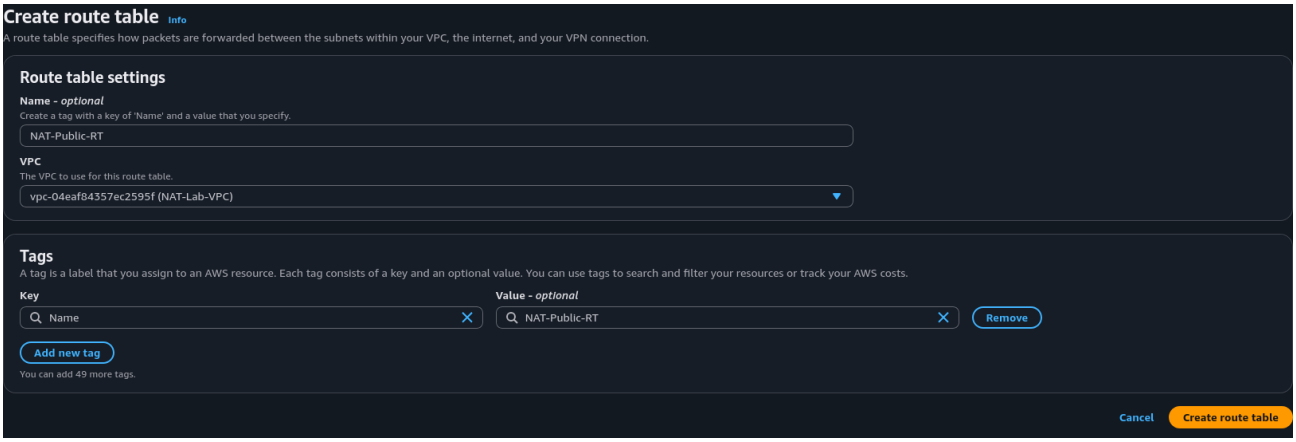


Figure 10: Creating `NAT-Public-RT`.
The route table belongs to the custom VPC and will have the local route plus a separately configured default route.

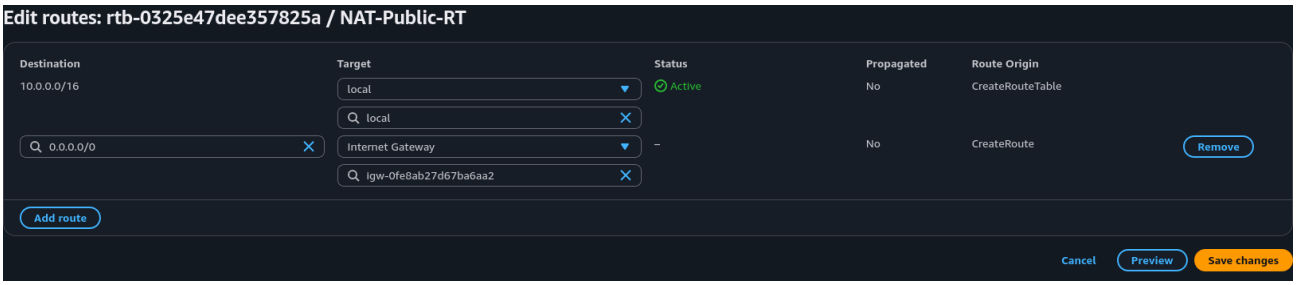


Figure 11: Adding the public IPv4 default route to the Internet Gateway.
`0.0.0.0/0` is the internet-bound route; `10.0.0.0/16` → `local` handles VPC-internal destinations.

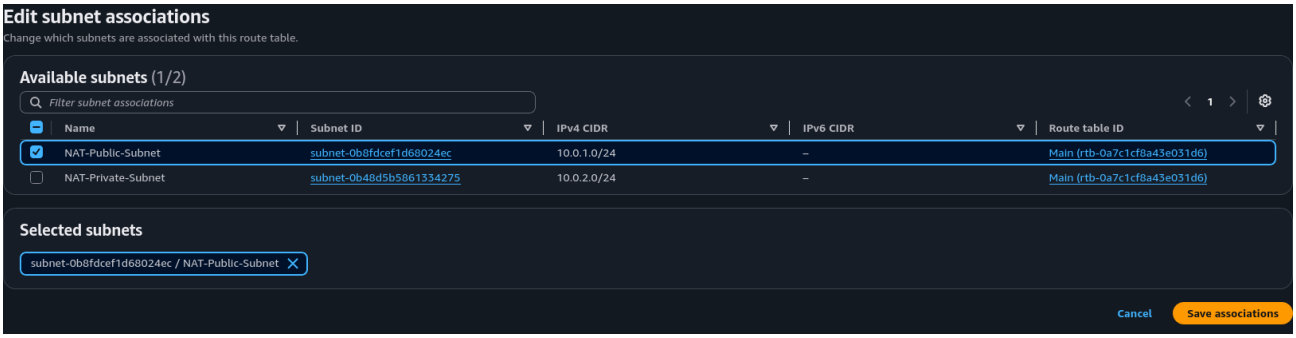


Figure 12: Associating `NAT-Public-Subnet` with `NAT-Public-RT`.
This screenshot concerns the public subnet association only.

10.2 Private route table

Create `NAT-Private-RT` and associate it with `NAT-Private-Subnet`. Before NAT, its intended routes contain only:

Destination	Target
`10.0.0.0/16`	`local`

The local route handles VPC-internal destinations but does not match Google's public address `8.8.8.8`; initially there is no route to that internet destination. This initial state is part of the learner-reported experimental setup. No pre-NAT private route-table screenshot was supplied.

11 Security group configuration

Security groups are stateful and attach to network interfaces. Responses to permitted outbound traffic are allowed back by connection tracking; the SSH inbound rules below do not themselves allow ICMP inbound initiation.

11.1 Public security group: `NAT-Public-SG`

Configure inbound SSH on TCP port 22 from the learner's current public IP, and retain the default outbound allow-all rule shown. Limiting the source is safer than exposing SSH to all IPv4 addresses (`0.0.0.0/0`). If the learner's public IP changes, update the rule when SSH is needed. The ping test is outbound ICMP, not SSH.

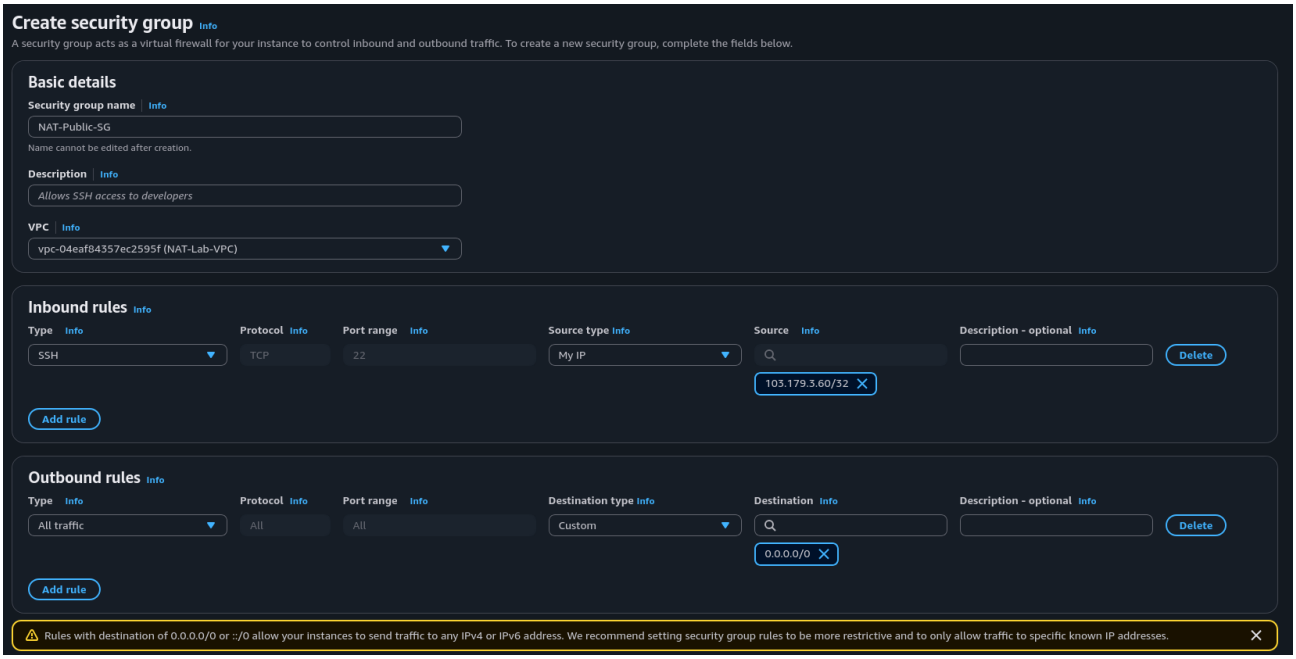


Figure 13: `NAT-Public-SG`: restricted SSH ingress and default outbound rule.
Notice TCP 22 uses the learner's public IP as its source, while outbound traffic is allowed by the visible default rule.

11.2 Private security group: `NAT-Private-SG`

Configure inbound SSH on TCP port 22 with `NAT-Public-SG` as source and retain default outbound allow-all. A security-group reference identifies interfaces associated with that group; it is not a fixed IP. This may support private-instance administration from the public instance, but it is not needed for NAT. The experiment did not demonstrate public-to-private SSH.

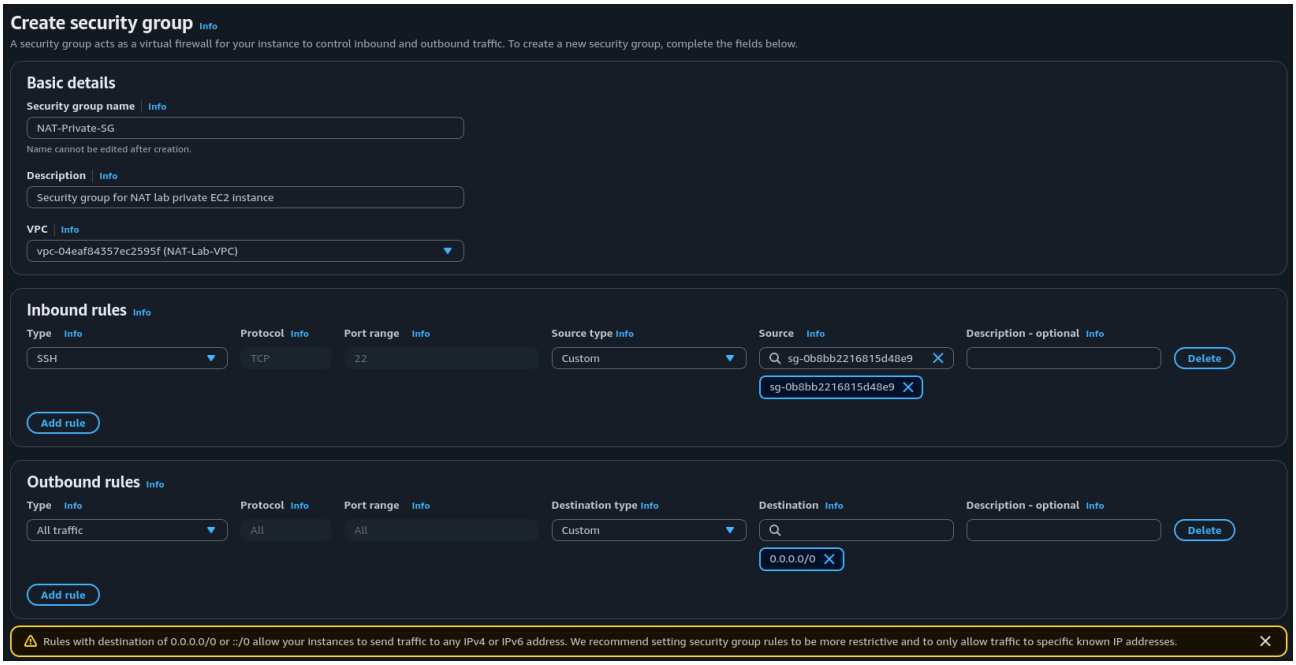


Figure 14: `NAT-Private-SG`: SSH ingress sourced from `NAT-Public-SG`.
This is a potential administration rule, not proof of SSH access or a NAT prerequisite. The screenshot also shows default outbound allow-all.

12 EC2 instance deployment

Launch `NAT-Public-EC2` in `NAT-Public-Subnet` with public IPv4 assignment enabled and `NAT-Public-SG` selected. The screenshot documents this visible network configuration. No AMI, instance type, instance ID, or IP address is inferred because those details are not specified in the available evidence.

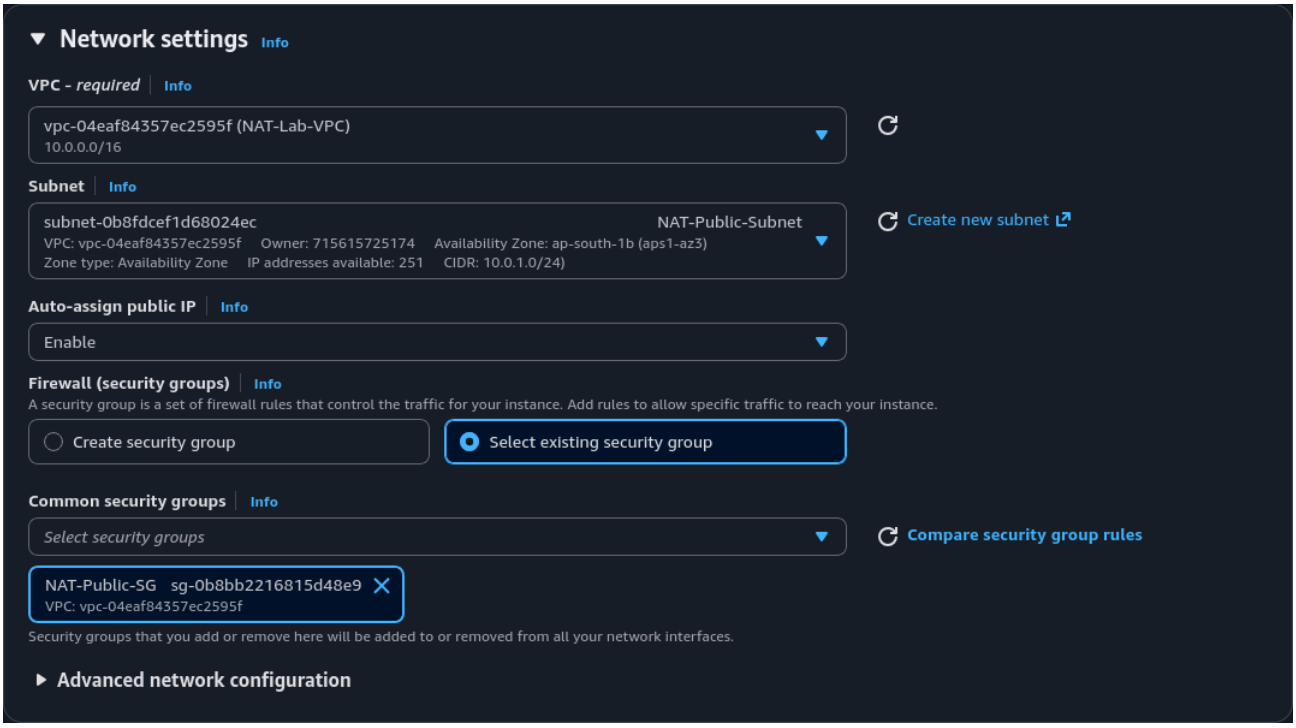


Figure 15: Public EC2 launch network configuration.
Notice the custom VPC, public subnet, public IPv4 assignment, and public security group. This is the public launch configuration only.

The learner also launched `NAT-Private-EC2` in `NAT-Private-Subnet` without a public IPv4 address. No

private-instance launch screenshot was supplied. Both instances were reported as running during testing. The intended difference is public IPv4 plus IGW routing for the public instance versus private addressing and initially no internet default route for the private instance.

13 Baseline connectivity tests before NAT

The same command was used on both instances to hold the destination and method constant:

```
ping -c 4 8.8.8.8
```

`ping` sends ICMP Echo Requests and displays replies; `-c 4` sends four requests and stops; `8.8.8.8` is Google's public DNS IPv4 address. ICMP does not use a TCP or UDP port.

Reported experimental outcomes (not terminal evidence in the supplied screenshots):

Instance	Test	Before NAT
`NAT-Public-EC2`	`ping -c 4 8.8.8.8`	Successful
`NAT-Private-EC2`	`ping -c 4 8.8.8.8`	Unsuccessful

The public result is expected with an IGW default route, public IPv4 connectivity, and permitted traffic. The private instance had no internet default route, so its route table could not direct the packet toward the public internet. A failed ping alone does not prove all internet protocols are unavailable: ICMP may be blocked or a destination may not respond. Here, the missing private default route is the documented routing explanation. These outcomes were learner-reported; no terminal output is fabricated or attributed to a console image.

14 NAT Gateway creation

Create the public NAT Gateway named `NAT-Lab-Gateway` in `NAT-Public-Subnet`, select public connectivity, and allocate or associate an Elastic IP. The gateway needs an internet path through the public subnet's route table and IGW. Its Elastic IP supplies a stable public IPv4 address for its internet-facing side; it is not assigned to the private EC2 instance.

A public NAT Gateway lets eligible private IPv4 resources initiate outbound connections without public IPv4 addresses. It does not make the private subnet public or provide arbitrary unsolicited inbound connectivity. The supplied creation form shows *Regional* selected. The learner later reported completing the experiment with a *Zonal* NAT Gateway. Thus, the form is not evidence of the final gateway's availability mode; Zonal is documented only as learner-reported.

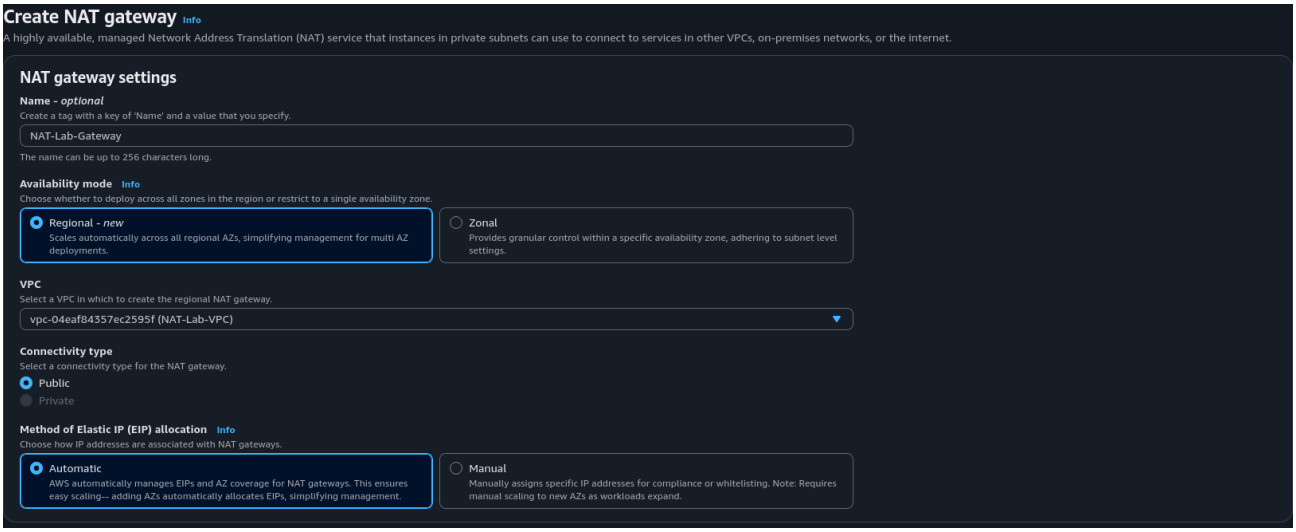


Figure 16: Public NAT Gateway creation form.
The form documents public connectivity and Elastic IP allocation options. It shows Regional selected at form time and does not establish final state or availability mode.

15 Private route table update

Once NAT is available, add the private table's default route:

`0.0.0.0/0` → `NAT-Lab-Gateway`

The supplied route screenshot shows NAT Gateway ID `nat-0bdd0403d6030b5d3` as target. The resulting routes are:

Destination	Target	Meaning
`10.0.0.0/16`	`local`	VPC-internal traffic
`0.0.0.0/0`	NAT Gateway	Other IPv4 destinations use NAT

Route tables select next hops; creating NAT does not automatically change private-subnet routing. This default route is essential.

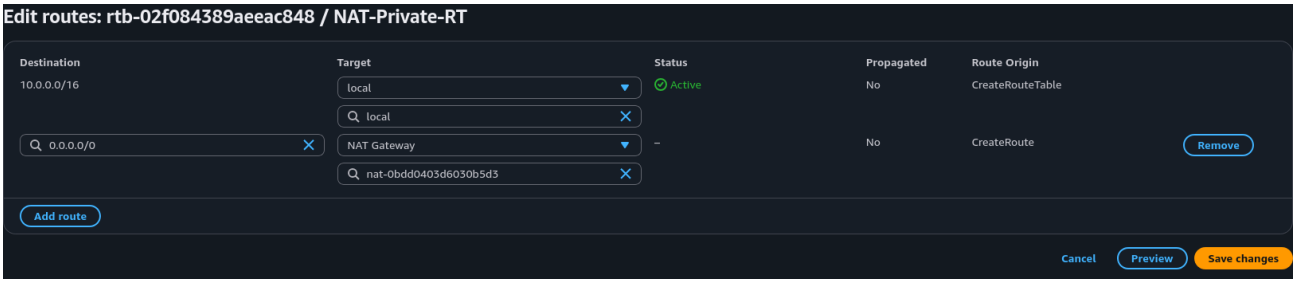


Figure 17: Private route table default route targeting NAT Gateway `nat-0bdd0403d6030b5d3`.
The screenshot shows the critical `0.0.0.0/0` NAT target and preserves `10.0.0.0/16` → `local`.

16 Connectivity test after NAT

After configuring the private default route, the learner repeated `ping -c 4 8.8.8.8` from `NAT-Private-EC2` and reported success. No terminal output or terminal screenshot was supplied, so this is an observed result reported by the learner, not screenshot-verified output.

Instance	Before NAT	After NAT
`NAT-Public-EC2`	Ping succeeded	No post-NAT result supplied
`NAT-Private-EC2`	Ping failed	Ping succeeded

The private outbound path is:

1. Private EC2 sends an ICMP Echo Request to `8.8.8.8`.
2. `NAT-Private-RT` matches `0.0.0.0/0` and directs the packet to NAT.
3. The public NAT Gateway translates the private source address for outbound IPv4 communication.
4. NAT forwards toward the internet through the public subnet's route to the IGW.
5. The reply returns through the established translation state and is forwarded to the private instance.

Successful ping depends on the route, NAT availability, permitted traffic, and the destination responding. NAT does not assign the private instance a public IPv4 address.

17 Results and analysis

The reported before-and-after outcomes demonstrate:

- Public internet connectivity used the public subnet's IGW route and public IPv4 configuration.
- Initially, the private instance could not reach the public test address because it had no internet default route.
- A public NAT Gateway plus the private default route enabled reported outbound connectivity without a public IPv4 address on the private instance.
- Creating NAT alone is insufficient; the private route table must target it, and NAT's public-subnet path must work.
- Route tables choose paths; security groups control permitted traffic. They have different jobs.
- NAT did not turn the private instance into a public instance.

Before NAT

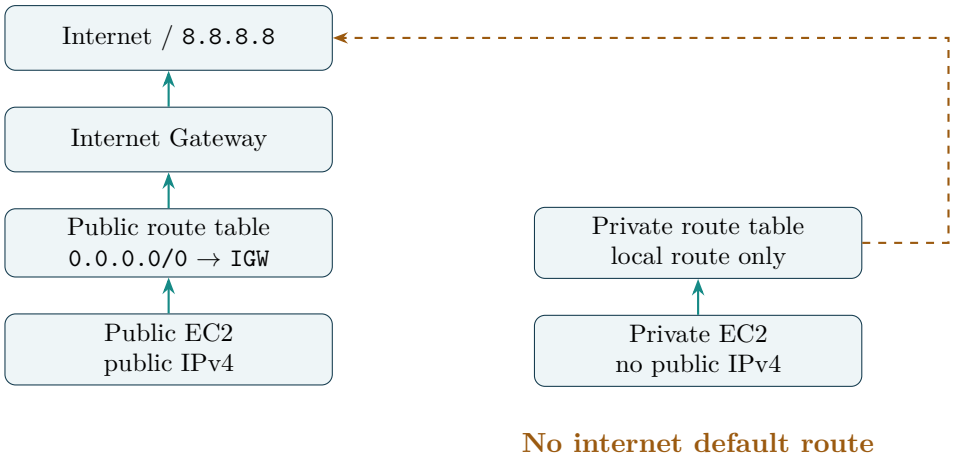


Figure 18: Before NAT: the public path reaches the IGW; the private route table has only its local route and no internet default route.

After NAT

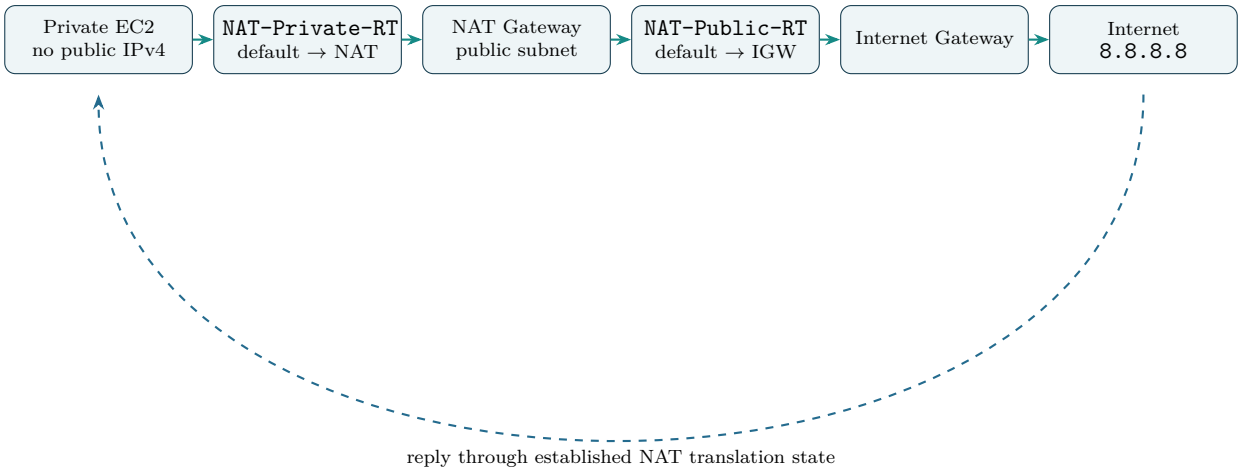


Figure 19: After NAT: route tables select successive next hops, and matching response traffic returns through the NAT translation state.

18 Troubleshooting and technical observations

Symptom	Possible cause	Check
Public instance cannot ping `8.8.8.8`	Wrong route, missing public IPv4, blocked ICMP	Public route table, instance addressing, security group, network ACL
Private instance cannot ping before NAT	Expected baseline result	Private route table has no internet default route
Private instance cannot ping after NAT	NAT unavailable, wrong target, public route issue, blocked ICMP	NAT state, both route tables, security groups, network ACLs
NAT Gateway cannot be created	Invalid subnet or missing required allocation/configuration	VPC, subnet, NAT settings, console error
SSH stops after public IP changes	SSH rule contains old source IP	Update source or use an appropriate managed access method
Route shows `blackhole`	Target deleted or unavailable	Check target resource and route table

Security groups are stateful; network ACLs are stateless, so both directions must be allowed as needed. Ping uses ICMP rather than a TCP/UDP port; checking SSH port 22 does not establish whether ICMP is permitted. No HTTPS test result is claimed.

19 Cleanup and cost considerations

After collecting evidence, remove resources in dependency-aware order. AWS may require dependent routes, associations, network interfaces, or gateways to be removed first. This is a recommended procedure, not a claim that cleanup occurred:

1. Delete the NAT Gateway and wait for deletion to complete.
2. Release its Elastic IP if no longer needed.
3. Remove the private route targeting NAT if it remains.
4. Terminate both lab EC2 instances.

5. Delete lab security groups after no network interfaces depend on them.
6. Delete custom route tables after resolving subnet associations.
7. Detach and delete the Internet Gateway.
8. Delete both subnets.
9. Delete `NAT-Lab-VPC`.

NAT Gateway may incur ongoing hourly and data-processing charges while provisioned. Elastic IP/public IPv4 charges may also apply depending on current pricing and allocation status. Do not assume every resource is Free Tier eligible. Review the AWS Billing console after the lab and verify no billable resources remain.

20 Final verification checklist

☐ Custom VPC created with planned IPv4 CIDR.

☐ Public and private subnets created without overlap.

☐ Internet Gateway attached to VPC.

☐ Public route table has default route to IGW.

☐ Public subnet associated with public route table.

☐ Security groups configured.

☐ Public and private EC2 instances launched.

☐ Public instance pinged `8.8.8.8` before NAT.

☐ Private instance failed baseline ping before NAT.

☐ Public NAT Gateway created.

☐ Private route table updated with NAT default route.

☐ Private instance successfully pinged `8.8.8.8` after NAT.

☐ Lab resources reviewed for cleanup.

☐ Billing reviewed for remaining resources.

21 Quick reference

Item	Value
Region	Asia Pacific (Mumbai), `ap-south-1`
VPC	`NAT-Lab-VPC`
VPC CIDR	`10.0.0.0/16`
Public subnet	`NAT-Public-Subnet`, `10.0.1.0/24`
Private subnet	`NAT-Private-Subnet`, `10.0.2.0/24`
Internet Gateway	`NAT-Lab-IGW`
Public route table	`NAT-Public-RT`
Private route table	`NAT-Private-RT`
NAT Gateway	`NAT-Lab-Gateway`
Public EC2	`NAT-Public-EC2`
Private EC2	`NAT-Private-EC2`
Public default route	`0.0.0.0/0` → `Internet Gateway`
Private default route after NAT	`0.0.0.0/0` → `NAT Gateway`
Test command	`ping -c 4 8.8.8.8`

22 Glossary

VPC Isolated virtual network in an AWS Region.

CIDR Notation defining an IP network range and prefix length.

Subnet VPC address range associated with one Availability Zone.

Public subnet
Subnet whose route table has an IGW route; direct internet IPv4 also needs public addressing and permitted traffic.

Private subnet
Subnet without a direct IGW route; outbound access may use NAT.

Route table
Rules selecting a next hop for destination prefixes.

Local route
VPC route enabling communication among VPC addresses.

Default route
Catch-all route when no more-specific route matches; IPv4 `0.0.0.0/0`.

Internet Gateway
VPC attachment serving as a target for internet routes.

NAT Gateway
Managed address translation for outbound connections from private IPv4 resources.

Elastic IP Static public IPv4 address allocated to an AWS account for supported resources.

EC2 AWS virtual compute instances service.

IAM AWS identity, policy, and permission service.

Security group
Stateful virtual firewall for network interfaces.

Network ACL
Stateless subnet-level network traffic filter.

ICMP Internet Control Message Protocol, used by IPv4 diagnostics such as ping.

Echo Request
ICMP message asking a destination to respond.

Echo Reply
ICMP response to an Echo Request.

Public IPv4 address
Internet-routable IPv4 address associated with an AWS resource.

Private IPv4 address
Address used inside private networks, not directly internet-routable.

Availability Zone
Isolated location within an AWS Region.

23 References

Official AWS documentation and pricing references (accessed October 27, 2026):

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